Day 35 Lab — Security Log Analysis & Alert Detection

BASIC COMMANDS

View the log: cat day35_auth.log (or: less day35_auth.log)

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ cat day35_auth.log
2026-10-09T10:00:01+05:30 | LOGIN_SUCCESS | user=student01 | source=192.0.2.10 | result=success
2026-10-09T10:00:45+05:30 | LOGIN_SUCCESS | user=student02 | source=192.0.2.14 | result=success
2026-10-09T10:02:12+05:30 | LOGIN_SUCCESS | user=student03 | source=192.0.2.18 | result=success
2026-10-09T10:03:07+05:30 | LOGOUT | user=student02 | source=192.0.2.14 | result=success
2026-10-09T10:04:33+05:30 | LOGIN_FAILED | user=student04 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:05:10+05:30 | LOGIN_SUCCESS | user=student05 | source=192.0.2.22 | result=success
2026-10-09T10:06:41+05:30 | LOGIN_SUCCESS | user=student01 | source=192.0.2.10 | result=success
2026-10-09T10:08:02+05:30 | PASSWORD_RESET_REQUEST | user=student04 | source=192.0.2.31 | result=email_sent
2026-10-09T10:09:18+05:30 | LOGIN_SUCCESS | user=student06 | source=192.0.2.40 | result=success
2026-10-09T10:10:00+05:30 | LOGIN_FAILED | user=student05 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:11:24+05:30 | LOGIN_SUCCESS | user=student07 | source=192.0.2.44 | result=success
2026-10-09T10:12:55+05:30 | LOGOUT | user=student03 | source=192.0.2.18 | result=success
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:20:14+05:30 | ACCESS_DENIED | user=student04 | source=192.0.2.31 | result=insufficient_permissions
```

Filter failed logins: grep "LOGIN_FAILED" day35_auth.log

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log
2026-10-09T10:04:33+05:30 | LOGIN_FAILED | user=student04 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:10:00+05:30 | LOGIN_FAILED | user=student05 | source=198.51.100.77 | result=invalid_credentials
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:27:44+05:30 | LOGIN_FAILED | user=student06 | source=192.0.2.40 | result=invalid_credentials
```

Count failed logins: grep "LOGIN_FAILED" day35_auth.log | wc -l

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | wc -l
16
```

Filter suspicious source: grep "203.0.113.25" day35_auth.log

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "203.0.113.25" day35_auth.log
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
```

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Failures from that source: grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25"

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25"
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials
```

LAB TASKS

Task 1: Open day35_auth.log and identify the available fields (Timestamp, Event Type, User, Source, Result).

Task 2: Count the total events

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ wc -l day35_auth.log
32 day35_auth.log
```

Task 3: Count the LOGIN_FAILED events.

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | wc -l
16
```

Task 4: Identify which source produced the highest suspicious failure pattern.

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | awk -F'|' '{print $3}' | sort | uniq -c | sort -nr
      4  user=student01
      2  user=student06
      2  user=student05
      2  user=student04
      2  user=student03
      2  user=student02
      2  user=admin_trainee
```

Task 5: Record the first suspicious timestamp, last suspicious timestamp and the approximate number of failures.

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```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25"
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:20+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:14:41+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:03+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:29+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:15:51+05:30 | LOGIN_FAILED | user=student02 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:12+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:16:40+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:05+05:30 | LOGIN_FAILED | user=admin_trainee | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:31+05:30 | LOGIN_FAILED | user=student04 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:17:58+05:30 | LOGIN_FAILED | user=student05 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:22+05:30 | LOGIN_FAILED | user=student03 | source=203.0.113.25 | result=invalid_credentials
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials

┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | head -1
2026-10-09T10:14:02+05:30 | LOGIN_FAILED | user=student01 | source=203.0.113.25 | result=invalid_credentials

┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | tail -1
2026-10-09T10:18:55+05:30 | LOGIN_FAILED | user=student06 | source=203.0.113.25 | result=invalid_credentials

┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep "LOGIN_FAILED" day35_auth.log | grep "203.0.113.25" | wc -l
13
```

Task 6: Determine whether this pattern would justify a security alert — and explain why.

Yes. This pattern justifies a security alert because:

1. The source IP 203.0.113.25 generated 13 failed login attempts within approximately five minutes.
2. Multiple user accounts were targeted, which may indicate an automated password-guessing or password-spraying attempt.
3. The activity could indicate an attempt to gain unauthorized access to user accounts.

Task 7: Design a simple alert rule: IF ___ WITHIN ___ THEN ___ SEVERITY ___.

IF more than 5 failed login attempts from the same source IP WITHIN 5 minutes THEN generate a security alert and notify the security team. SEVERITY HIGH.

Task 8: Review the log for sensitive information — confirm it contains no passwords, tokens or MFA secrets, and explain why these values should not normally be logged.

```
┌──(kirthikaa㉿kali)-[~/Desktop]
└─$ grep -Ein 'password|passwd|token|secret|api_key|session_id|mfa_code|otp' day35_auth.log
8:2026-10-09T10:08:02+05:30 | PASSWORD_RESET_REQUEST | user=student04 | source=192.0.2.31 | result=email_sent
```

- The authentication log contains timestamps, event types, usernames, source IP addresses and authentication results. It does not visibly contain plaintext passwords, authentication tokens or MFA secrets.
- Sensitive information should not be logged because attackers who gain access to log files could steal credentials and use them to compromise accounts.
- Applications should avoid recording passwords, session tokens, API keys and MFA secrets. Logs should contain only the information necessary for security monitoring and investigation.

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Task 9: Write a short incident summary: what happened, when, which source, which fictional accounts were targeted, and the recommended next action.

On 9 October 2026, suspicious login activity was detected from the fictional IP address 203.0.113.25. The source generated 13 failed login attempts between 10:14:02 and 10:18:55, targeting seven accounts: student01, student02, student03, student04, student05, student06 and admin_trainee.

This activity may indicate an automated password-guessing or password-spraying attempt. A HIGH severity alert is recommended. The security team should investigate the source, monitor the targeted accounts, review authentication activity and enforce MFA.